Downloaded from http://journals.lww.com/pidj by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC1y0abggQZXdtwnfKZBYtws= on 08/11/2023

(Fig. C, Supplemental Digital Content 1, http://links.lww.com/INF/B601) whereas increased intraorbital contrast enhancement was seen on T1-weighted images indicating an inflammatory process. In view of the bilateral ethmoid sinusitis, a deep nasal swab was taken. The patient was treated with dexamethason, mannitol and low-dose unfractionated heparin. A follow-up unenhanced MRI on day +7 showed additional bilateral infarction in the territory of the middle cerebral artery and extensive cerebral edema (Fig. D, Supplemental Digital Content 1, http://links.lww.com/INF/B601). The patient died on day +8 after peripheral blood stem cell transplantation. Cultures from the deep nasal swab taken on day +7 were positive for *L. corymbifera*. Galactomannan blood tests were negative. Parents denied autopsy.

DISCUSSION

ROCM is a rare fulminant fungal infection that can lead to devastating consequences in immunocompromised neutropenic patients. Lack of phagocytes in patients with hematologic malignancies after chemotherapy or stem cell transplantation is a major risk factor for the development of ROCM.[2,7,8] Mortality still ranges from 40% to 80% and is increased when intracranial dissemination occurs.[3,8,9] Initial clinical symptoms are nonspecific and comprise fever, rhinorrhea, headache and sinus pain.[2] Progression of the infection to the orbit, the optic nerve and the ophthalmic vessels leads to facial or periorbital swelling, proptosis, eyepain and loss of vision.[2] Invasion through the ethmoid sinus may lead to cavernous sinus thrombosis and consequent affection of other cranial nerves such as III, IV and VI and altered state of consciousness.[2,3] Our patient initially developed bilateral peripheral oculomotor nerve palsy (areactive mydriasis and ptosis with normal Glasgow Coma Scale) followed by complete bilateral ophthalmoplegia and loss of vision. This association is known as the orbital apex syndrome and can occur along orbital infections, neoplasms, injuries or vascular conditions. The hallmark of ROCM is the rapidity of its dissemination. In our case, only a few hours passed from the onset of unilateral mydriasis to the full picture of the orbital apex syndrome and comatose state.

Diagnosis or ROCM rests on positive fungal culture in swabs or tissue biopsies and imaging to illustrate the site of infection.[2,3] Computerized tomography scanning shows mucosal thickening of the nasal cavity and paranasal sinuses, enhancement of soft tissue masses or bony destruction.[3,6] MRI is more sensitive concerning intraorbital and intracerebral involvement. Intraorbital invasion is characterized by thickening of the medial rectus muscle and inflammatory orbital fat changes appearing hyperintense on T2-weighted images, in particular at the orbital apex.[3,6] Inflamed mucosa shows high signal intensity on T2-weighted images, whereas sinus masses typically display low signal intensity due to high concentrations of metals such as iron, magnesium or manganese.[6] Radiologic signs of CST can be more difficult to assess in the early phase and clinical signs often precede imaging findings.[2] On T1-weighted images, CST appears isointense with enhancement, whereas on T2-weighted images, it appears hyperintense.[6] Invasion of the internal carotid arteries can be assessed by magnetic resonance angiography.

In our case, computerized tomography scans were normal, whereas MRI performed on the following day revealed ethmoid sinusitis, intraorbital inflammatory alterations and thrombosis of the internal carotid artery at the level of the cavernous sinus.

The outcome of ROCM is dependent on several factors including underlying disease, extent of the infection, therapeutic delay and treatment modalities.[9] In our patient, recurrent swabs had been negative, and only a deep nose swab finally allowed to cultivate *L. corymbifera*.

L. corymbifera is a microorganism with high propensity for angioinvasion and tissue necrosis. In our case, most probably invasion and retrograde inflammatory occlusion of both internal carotid arteries caused bilateral infarctions of the frontal and middle cerebral arterial territories with fatal brain swelling. This assumption is based on the clinical picture of the orbital apex syndrome and the radiological findings of ethmoid sinusitis and orbital involvement. However, carotid artery dissection remains a potential differential diagnosis.

In summary, this case illustrates that invasive ROCM leaves an extremely short time window for therapeutic options. Neutropenic patients presenting with clinical signs of sinusitis or visual disturbances should always undergo a rapid diagnostic work-up for ROCM.[2] Treatment should be initiated immediately, irrespective of imaging findings until results of microbiological testing are available.[10] Administration of intravenous liposomal amphotericin B after diagnostic biopsies and surgical debridement is considered the treatment of choice.[10] Furthermore, control of underlying causes such as immunodeficiency and neutropenia is of utmost importance.[10]

REFERENCES

1. Roden MM, Zaoutis TE, Buchanan WL, et al. Epidemiology and outcome of zygomycosis: a review of 929 reported cases. *Clin Infect Dis*. 2005;41:634–653.
2. Gamaletsou MN, Sipsas NV, Roilides E, et al. Rhino-orbital-cerebral mucormycosis. *Curr Infect Dis Rep*. 2012;14:423–434.
3. Koc Z, Koc F, Yerdelen D, et al. Rhino-orbital-cerebral mucormycosis with different cerebral involvements: infarct, hemorrhage, and ophthalmoplegia. *Int J Neurosci*. 2007;117:1677–1690.
4. Margo CE, Linden C, Strickland-Marmol LB, et al. Rhinocerebral mucormycosis with perineural spread. *Ophthal Plast Reconstr Surg*. 2007;23:326–327.
5. Thajeb P, Thajeb T, Dai D. Fatal strokes in patients with rhino-orbito-cerebral mucormycosis and associated vasculopathy. *Scand J Infect Dis*. 2004;36:643–648.
6. Aribandi M, McCoy VA, Bazan C 3rd. Imaging features of invasive and noninvasive fungal sinusitis: a review. *Radiographics*. 2007;27:1283–1296.
7. Ibrahim AS, Spellberg B, Walsh TJ, et al. Pathogenesis of mucormycosis. *Clin Infect Dis*. 2012;54(suppl 1):S16–S22.
8. Dehority W, Willert J, Pong A. Zygomycetes infections in pediatric hematology oncology patients: a case series and review of the literature. *J Pediatr Hematol Oncol*. 2009;31:911–919.
9. Santos Gorjón P, Blanco Pérez P, Batuecas Caletrío A, et al. Rhino-orbito-cerebral mucormycosis, a retrospective study of 7 cases. *Acta Otorrinolaringol Esp*. 2010;61:48–53.
10. Skiada A, Lanternier F, Groll AH, et al.; European Conference on Infections in Leukemia. Diagnosis and treatment of mucormycosis in patients with hematological malignancies: guidelines from the 3rd European Conference on Infections in Leukemia (ECIL 3). *Haematologica*. 2013;98:492–504.

TRAUMATIC LUMBAR PUNCTURES IN INFANTS HOSPITALIZED IN THE NEONATAL INTENSIVE CARE UNIT

Lakshmi Srinivasan, MB BS, Samir S. Shah, MD, MSCE,† Soraya Abbasi, MD,‡ Michael A. Padula, MD, MBI,* and Mary C. Harris, MD**

Abstract: Traumatic lumbar punctures occur frequently in the neonatal intensive care unit, making the interpretation of cerebrospinal fluid values difficult. We report correction factors for cerebrospinal fluid protein and white blood cells in the face of red blood cell contamination. These correction factors should facilitate the diagnosis of bacterial meningitis in high-risk hospitalized infants.

Key Words: cerebrospinal fluid, infant, neonate, diagnosis, meningitis, lumbar puncture, traumatic, reference values

© 2013 Lippincott Williams & Wilkins

Downloaded from http://journals.lww.com/pidj by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC1y0abggQZXdtwnfKZBYtws= on 08/11/2023

Accepted for publication April 23, 2013.
From the *Department of Pediatrics, The Children's Hospital of Philadelphia, Philadelphia, PA; †Division of Hospital Medicine, Cincinnati Children's Hospital Medical Center, Cincinnati, OH; and ‡Division of Newborn Care, Pennsylvania Hospital, Philadelphia, PA.
This study was supported by Clinical and Translational Research Center grant number UL1-RR-024134. M.C.H. received support from the University Research Foundation and the Foerderer Murray Award. S.S.S. received support from the Robert Wood Johnson Foundation's Physician Faculty Scholar Program. The content of this article is solely the responsibility of the authors and does not represent the official views of the funding agencies mentioned above. The authors have no other funding or conflicts of interest to disclose.
Address for correspondence: Lakshmi Srinivasan, MB BS, The Children's Hospital of Philadelphia, Room 2NW17, 34th Street and Civic Center Boulevard, Philadelphia, PA 19104. E-mail: srinivasanl@email.chop.edu.
Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's website (www.pidj.com).
Copyright © 2013 by Lippincott Williams & Wilkins
ISSN: 0891-3668/13/3210-1151
DOI: 10.1097/INF.0b013e31829862b7

Neonatal meningitis is a rare but potentially devastating condition.[1,2] Infants in the neonatal intensive care unit (NICU) often receive lumbar punctures (LPs) as part of the "sepsis workup" to rule out meningitis. However, LPs are often delayed due to cardiorespiratory instability or other concerns, and antibiotics are administered empirically.[3] In these situations, the accuracy of the cerebrospinal fluid (CSF) culture is diminished, and clinicians rely on the results of other CSF values, especially the CSF white blood cell (WBC) count, to make decisions regarding the likelihood of bacterial meningitis.[4,5]

Traumatic LPs occur in 35% to 46% of infants, making interpretation of CSF cultures difficult.[6] Conventional wisdom regarding correction factors is based on the premise that the ratio of WBCs to red blood cells (RBCs) in blood-contaminated CSF is similar to peripheral blood (around 1:500). This correction rule is used to calculate the CSF WBC count attributable to contamination of CSF by blood.[7] Others have suggested an "observed:predicted ratio," the observed CSF WBC count compared with that predicted based on the peripheral blood WBC:RBC ratio.[8] Clinicians often use these rules to make decisions regarding treatment; if the actual number of CSF WBC exceeds that predicted, antibiotic therapy may be altered or prolonged to treat "presumed meningitis."

Although several studies have evaluated correction rules in adults and older children, only a few have examined their validity in infants.[9] A previous multicenter retrospective analysis utilizing a large neonatal administrative database found that adjustment of WBC counts in traumatic taps using correction rules led to decreased sensitivity with a slight increase in specificity.[6] A recent study of traumatic LPs in the emergency department presented a useful prediction rule for the change in CSF protein with increasing CSF RBC.[10]

The aim of our study was to quantify changes in CSF protein, WBC and glucose in the presence of elevated CSF RBC among hospitalized infants in the NICU.

MATERIALS AND METHODS

We performed a secondary analysis from a prospective observational study of hospitalized infants <6 months of age receiving LPs whose primary aim was to examine the impact of bacterial polymerase chain reaction on the diagnosis of bacterial meningitis. This study was conducted in 3 NICUs located in Philadelphia from March 1, 2008, to December 31, 2010. To quantify accurately changes in CSF values with increasing CSF RBC, we sequentially excluded infants with known potential causes of CSF pleocytosis or protein elevation (culture-proven or clinically presumed meningitis, viral meningitis, bacteremia, ventriculoperitoneal shunts, intraventricular hemorrhage, contaminants, unknown CSF culture results and outlier values [top 1%] for CSF WBC [>180 cells/mm^3], RBC [>100,000 cells/mm^3] and protein [>550 mg/dL]). Using linear regression, we determined the association of CSF protein and WBC with increasing CSF RBC. Predicted CSF WBCs were calculated as (peripheral WBC count/peripheral RBC count) × CSF RBC count. The observed:predicted CSF WBC ratio was calculated as measured CSF WBC/predicted CSF WBC. Multiple linear regression was used to assess timing of initiation of antibiotics and gestational age as additional factors that might impact CSF values.

TABLE 1. Demographic Data and CSF Values

Variable	Infants With Nontraumatic LPs (n = 365)	Infants With Traumatic LPs (n = 130)	*P*
Demographic data			
Postnatal age (d) Median (IQR)	3 (1–17)	2 (1–7)	0.003*
Gestational age (wk) Median (IQR)	38 (33–40)	37 (33–40)	0.776*
Birthweight (g) Median (IQR)	2853 (1891–3386)	2825 (1956–3400)	0.823*
Male gender n (%)	210 (58%)	74 (57%)	0.904†
Preterm (<37 wk) n (%)	159 (44%)	60 (46%)	0.609†
CSF values			
CSF WBC (cells/mm^3) Median (IQR)	3 (1–6)	6 (2–13)	<0.001*
CSF protein (mg/dL) Median (IQR)	85 (62–116)	106 (82–137)	<0.001*
CSF glucose (mg/dL) Median (IQR)	50 (43–60)	49 (43–59)	0.309*

*Wilcoxon rank sum test.
†χ^2 test.
IQR, interquartile range.

Study Definitions

We defined traumatic LPs as CSF RBC >1000 cells/mm^3. Proven meningitis was defined based on the presence of positive CSF cultures with organisms other than those regarded as contaminants. Clinically presumed meningitis was diagnosed based on the presence of abnormal CSF values as well as the decision to treat for meningitis, even in the presence of a negative CSF culture. Abnormal CSF values were defined as follows: elevated CSF WBC, defined as >22 CSF WBC if less than 28 days old, and >15 CSF WBC if more than 28 days old, and 1 of the following: positive CSF gram stain or elevated CSF protein (>120 gm/dL) or low CSF glucose (<20 mg/dL).[5]

RESULTS

One hundred thirty of 495 infants (26.3%) had traumatic LPs. These infants were slightly younger, but there were no significant differences in gestational age, birth weight or gender between groups (Table 1). Overall, 44% were preterm (gestational age <37 weeks). Seventy-eight percent of infants with traumatic LPs received antibiotics before the LP.

CSF WBCs were increased in infants with traumatic LPs compared with infants without traumatic LPs, as was CSF protein (Table 1). CSF glucose values were not significantly different between groups.

Downloaded from http://journals.lww.com/pidj by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC1y0abggQZXdtwnfKZBYtws= on 08/11/2023

CSF WBCs increased by 3 cells/mm^3 per 10,000 CSF RBCs (95% confidence interval [CI]: 2–4 cells/mm^3). When stratified by prematurity, CSF WBCs rose by 2 cells/mm^3 per 10,000 CSF RBCs in preterm infants (95% CI: 1–3 cells/mm^3) versus 4 cells/mm^3 per 10,000 CSF RBCs in term infants (95% CI: 2–6 cells/mm^3) (Fig. A, Supplemental Digital Content 1, http://links.lww.com/INF/B565). CSF protein increased by 1.3 mg/dL per 1000 CSF RBCs (95% CI: 0.9–1.8 mg/dL). In preterm infants, the rate of increase was 1.1 mg/dL per 1000 CSF RBCs (95% CI: 0.3–1.9 mg/dL) versus 1.5 mg/dL in term infants (95% CI: 1–2 mg/dL) (Fig. B, Supplemental Digital Content 1, http://links.lww.com/INF/B565).

Using multiple linear regression, timing of initiation of antibiotics had no significant effect on either CSF WBC or protein values. CSF protein decreased by 3 mg/dL for every additional week of gestation (P = 0.015). However, the relationship of CSF RBC values with CSF WBC and protein values remained significant in this model, of a magnitude similar to that found on simple linear regression (results not shown).

Observed:predicted CSF WBC ratios varied widely (Fig., Supplemental Digital Content 2, http://links.lww.com/INF/B566). In general the predicted CSF WBC was around 900 times the observed CSF WBC value.

DISCUSSION

Traumatic LPs are a frequent occurrence in the NICU and present a clinical dilemma when making decisions about treatment for bacterial meningitis.[6–8] Our study provides correction factors that can be applied to the NICU population to determine expected elevations of CSF WBCs and protein in the setting of high CSF RBC values.

In a large retrospective analysis, Greenberg et al[6] showed that the 1:500 CSF WBC:RBC rule and the observed:predicted ratio failed to improve the accuracy of the diagnosis of neonatal meningitis. In our carefully selected population of presumably uninfected infants, one would expect the observed:predicted CSF WBC ratio to be close to 1. However, we found predicted CSF WBC values that were several-fold higher than observed values, suggesting that this correction rule is of limited diagnostic value in this population. Our prospective study, therefore, reinforces the data from Greenberg and also provides an alternative correction rule for CSF WBC elevations in the setting of traumatic LPs—a rise of only 1 cell/mm^3 for every 3333 CSF RBCs.

Hines and colleagues[10] predicted a 1.9 mg/dL increase in CSF protein per 1000 CSF RBCs/mm^3 in traumatic LPs in neonates in the emergency department. Our finding of a 1.5 mg/dL rise in CSF protein per 1000 RBCs in term infants suggests that this rule is robust and may be generalizable to both NICU and emergency department settings. In addition, we provide a separate rule for preterm infants in the NICU, with a slightly lower rate of increase in CSF protein.

The strengths of our study include rigorous data collection as part of a prospective multisite study with few missing data. We applied stringent criteria to exclude other potential causes of CSF pleocytosis to determine the most accurate quantification of the relationship between CSF WBC and RBC. Our study provides correction rules for the NICU that validate and extend the results of a previous study in the emergency department.

This study has several limitations. First, we could not exclude the possibility of bacterial meningitis in infants receiving antibiotics before the LP. The inadvertent inclusion of infants with bacterial meningitis would cause us to overestimate the correction of CSF WBCs in the setting of a traumatic LP. However, we believe it is unlikely that infants with bacterial meningitis were included because we excluded infants with very high CSF WBC counts and those with possible bacterial meningitis. It is also unlikely that infants with bacterial meningitis had normal CSF protein and glucose values and the absence of bacteria on CSF gram stain. Secondly, we eliminated extreme outliers to prevent skewing of our results. Therefore, our results cannot be extrapolated to infants with grossly bloody LPs or extremely high CSF RBC counts. Furthermore, although we attempted to account for the effect of antibiotic pretreatment and gestational age using multivariate models, the relatively modest sample size of this study did not allow further stratified analyses. Lastly, changes in CSF RBCs appear to account for only a small proportion of the variability in CSF WBC and protein. So, although these rules are useful to predict changes in CSF values, additional factors are also likely to impact these measurements.

In conclusion, we provide correction rules for CSF WBC counts and protein in the setting of traumatic LPs. These findings may improve interpretation of CSF values and aid decision making regarding the diagnosis of bacterial meningitis in high-risk hospitalized infants.

REFERENCES

1. Stoll BJ, Hansen N, Fanaroff AA, et al. To tap or not to tap: high likelihood of meningitis without sepsis among very low birth weight infants. *Pediatrics*. 2004;113:1181–1186.
2. Adams-Chapman I, Stoll BJ. Neonatal infection and long-term neurodevelopmental outcome in the preterm infant. *Curr Opin Infect Dis*. 2006;19:290–297.
3. Srinivasan L, Shah SS, Padula MA, et al. Cerebrospinal fluid reference ranges in term and preterm infants in the neonatal intensive care unit. *J Pediatr*. 2012;161:729–734.
4. Kanegaye JT, Soliemanzadeh P, Bradley JS. Lumbar puncture in pediatric bacterial meningitis: defining the time interval for recovery of cerebrospinal fluid pathogens after parenteral antibiotic pretreatment. *Pediatrics*. 2001;108:1169–1174.
5. Garges HP, Moody MA, Cotten CM, et al. Neonatal meningitis: what is the correlation among cerebrospinal fluid cultures, blood cultures, and cerebrospinal fluid parameters? *Pediatrics*. 2006;117:1094–1100.
6. Greenberg RG, Smith PB, Cotten CM, et al. Traumatic lumbar punctures in neonates: test performance of the cerebrospinal fluid white blood cell count. *Pediatr Infect Dis J*. 2008;27:1047–1051.
7. Bonsu BK, Harper MB. Corrections for leukocytes and percent of neutrophils do not match observations in blood-contaminated cerebrospinal fluid and have no value over uncorrected cells for diagnosis. *Pediatr Infect Dis J*. 2006;25:8–11.
8. Bonadio WA, Smith DS, Goddard S, et al. Distinguishing cerebrospinal fluid abnormalities in children with bacterial meningitis and traumatic lumbar puncture. *J Infect Dis*. 1990;162:251–254.
9. Nigrovic LE, Shah SS, Neuman MI. Correction of cerebrospinal fluid protein for the presence of red blood cells in children with a traumatic lumbar puncture. *J Pediatr*. 2011;159:158–159.
10. Hines EM, Nigrovic LE, Neuman MI, et al. Adjustment of cerebrospinal fluid protein for red blood cells in neonates and young infants. *J Hosp Med*. 2012;7:325–328.

SEVERE PERTUSSIS IN NEWBORNS AND YOUNG VULNERABLE INFANTS

Gustavo Rocha, MD, Filipa Flôr-de-Lima, MD,* Paulo Soares, MD,*† Henrique Soares, MD,*† Susana Pissarra, MD,*† Elisa Proença, MD,‡ Paula Fernandes, MD,§ Conceição Quintas, MD,¶ Teresa Martins, MD,‖ Albina Silva, MD,** and Hercília Guimarães, PhD*†*

Abstract: A retrospective chart review of 18 nonvaccinated newborns and infants admitted to 6 intensive care units in the north of Portugal between 2007 and 2012 revealed a high rate of admissions in 2012 along with significant rates of severe pulmonary hypertension and mortality. Hyperleu-

 © 2013 Lippincott Williams & Wilkins